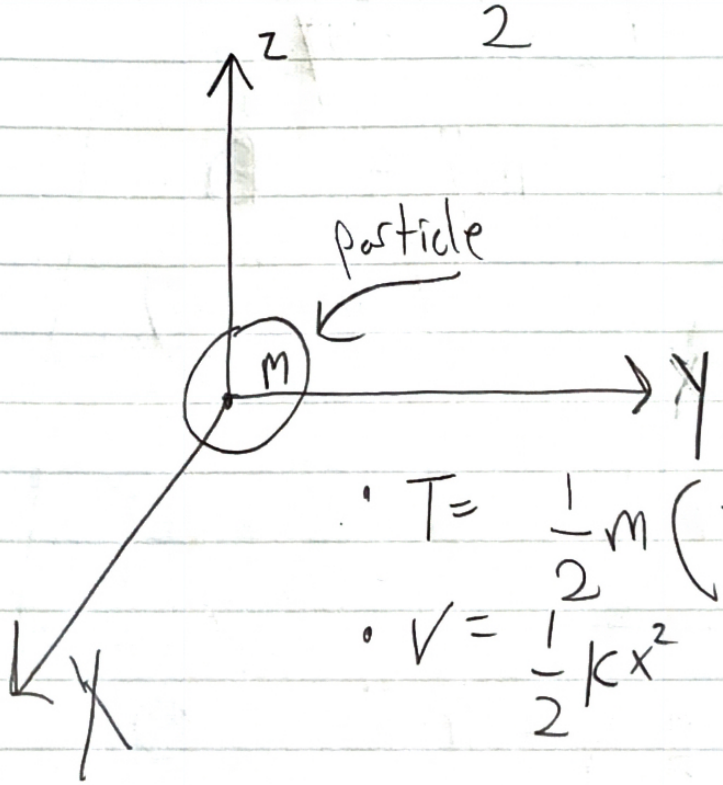
Homework #10

Parker Lewis

	hrs
prep	10
Prob #1	1
Prob #2	4
Prob #3	1
Total	16

Prob #1)

- $V = \frac{1}{2} k x^2$



- $T = \frac{1}{2} m \left(\dot{x}^2 + \dot{y}^2 + \dot{z}^2 \right)$
- $V = \frac{1}{2} k x^2$

$$L = T - V = \frac{1}{2} m\left(\dot{x}^2 + \dot{y}^2 + \dot{z}^2\right) - \frac{k}{2} x^2$$

$$\frac{\partial L}{\partial x} = kx \quad , \quad \frac{\partial L}{\partial \dot{x}} = m\dot{x}$$

$$\frac{\partial}{\partial t}\left(\frac{\partial L}{\partial \dot{x}}\right) - \frac{\partial L}{\partial x} = m\ddot{x} - kx = 0$$

(x-equation)

$$\frac{\partial L}{\partial \dot{y}} = m\dot{y} \quad , \quad \frac{\partial L}{\partial y} = 0$$

$$\frac{\partial}{\partial t}\left(\frac{\partial L}{\partial \dot{y}}\right) = m\ddot{y} = 0$$

(y-equation)

$$\frac{\partial L}{\partial \dot{z}} = m\dot{z} \quad , \quad \frac{\partial L}{\partial z} = 0$$

$$\frac{\partial}{\partial t}\left(\frac{\partial L}{\partial \dot{z}}\right) = m\ddot{z} = 0$$

(z-eqn)

X-equation:

- $m\ddot{x} - kx = 0$
- $\ddot{x} - \frac{k}{m}x = 0$, $\omega_x^2 = \frac{k}{m}$

$$\omega_x = \sqrt{\frac{k}{m}}$$

Y-equation:

- $m\ddot{y} = 0$

$$\omega_y = 0$$

Z-equation:

- $m\ddot{z} = 0$

$$\omega_z = 0$$

X-equation:

$$\ddot{X} - \frac{k}{m}X = 0$$

$$r^2 - \frac{k}{m} = 0$$

$$r = \pm\sqrt{\frac{k}{m}} = \pm\omega_x$$

$$X = e^{rt}$$

$$\dot{X} = re^{rt}$$

$$\ddot{X} = r^2 e^{rt}$$

$$X(t) = A\cos(\omega_x t) + B\sin(\omega_x t)$$

Initial condition)

$X(0) = X_0$ (displacement value)

$\dot{X}(0) = 0$

$X_0 = A\cos(0) + B\sin(0)$

$A = X_0$

$\dot{X}(t) = -A\omega_x \sin(\omega_x t) + B\omega_x \cos(\omega_x t)$

$B = 0, \quad 0 = B\cos(0) = B$

$$X(t) = X_0 \cos(\omega_x t)$$

y-equation:

- $m\ddot{y} = 0$
- $m\int \dot{y}\,dy = \int 0\,dt$ (antiderivative of 0 → constant term)
- $m\dot{y} = C_1$
- $m\int dy = \int C_1\,dt$
- $m\,y(t) = C_1 t + C_2$
- $\boxed{y(t) = \frac{C_1}{m}t + \frac{C_2}{m}}$

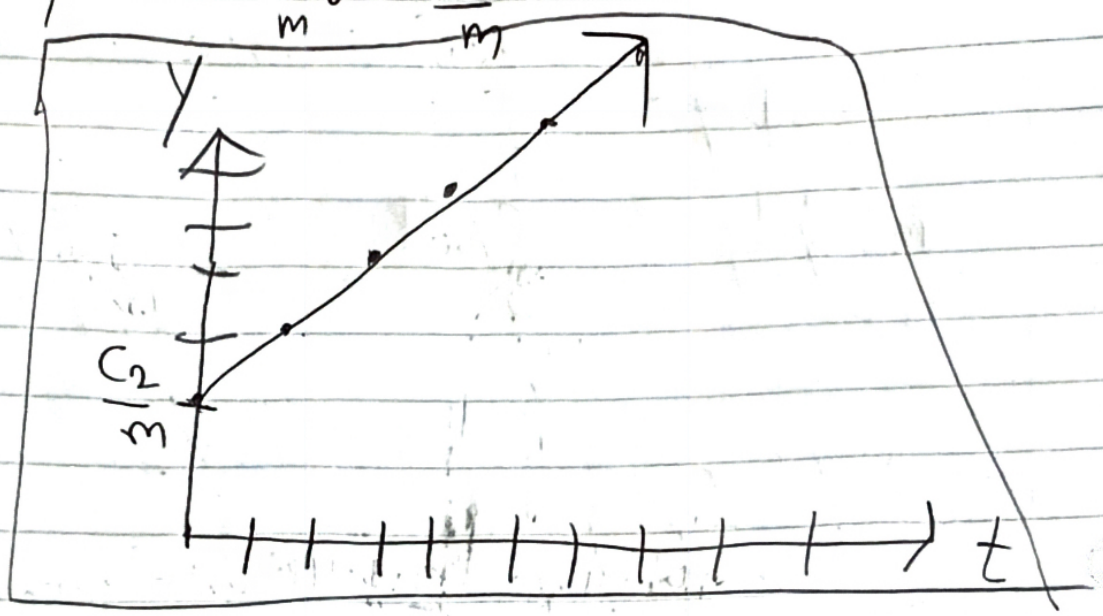
z-equation:

Same form as y-equation:

$$\boxed{z(t) = \frac{C_1}{m}t + \frac{C_2}{m}}$$

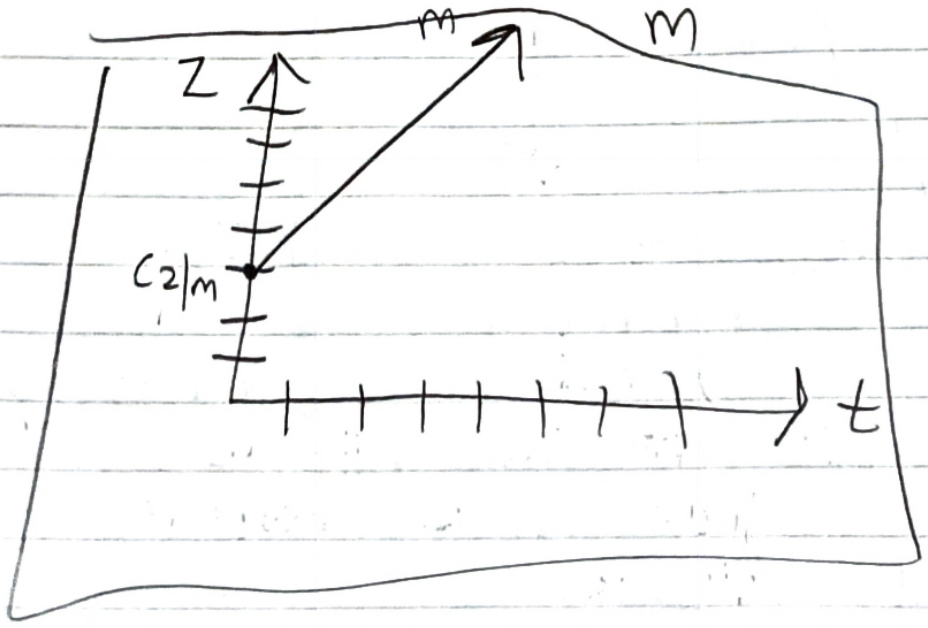
y-equation:

$$y(t) = \frac{c_1}{m} t + \frac{c_2}{m}$$



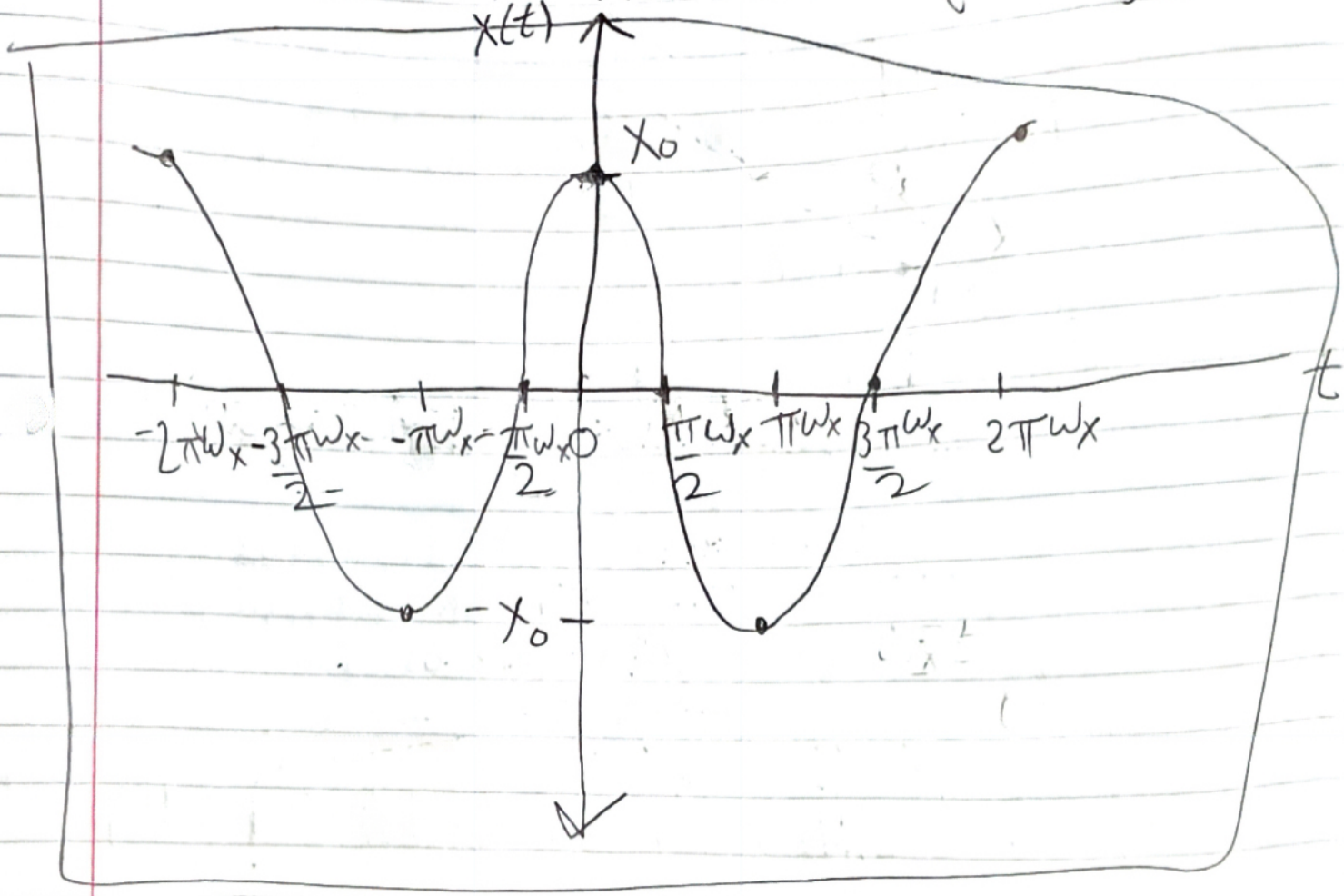
z-equation:

$$z(t) = \frac{c_1}{m} t + \frac{c_2}{m}$$



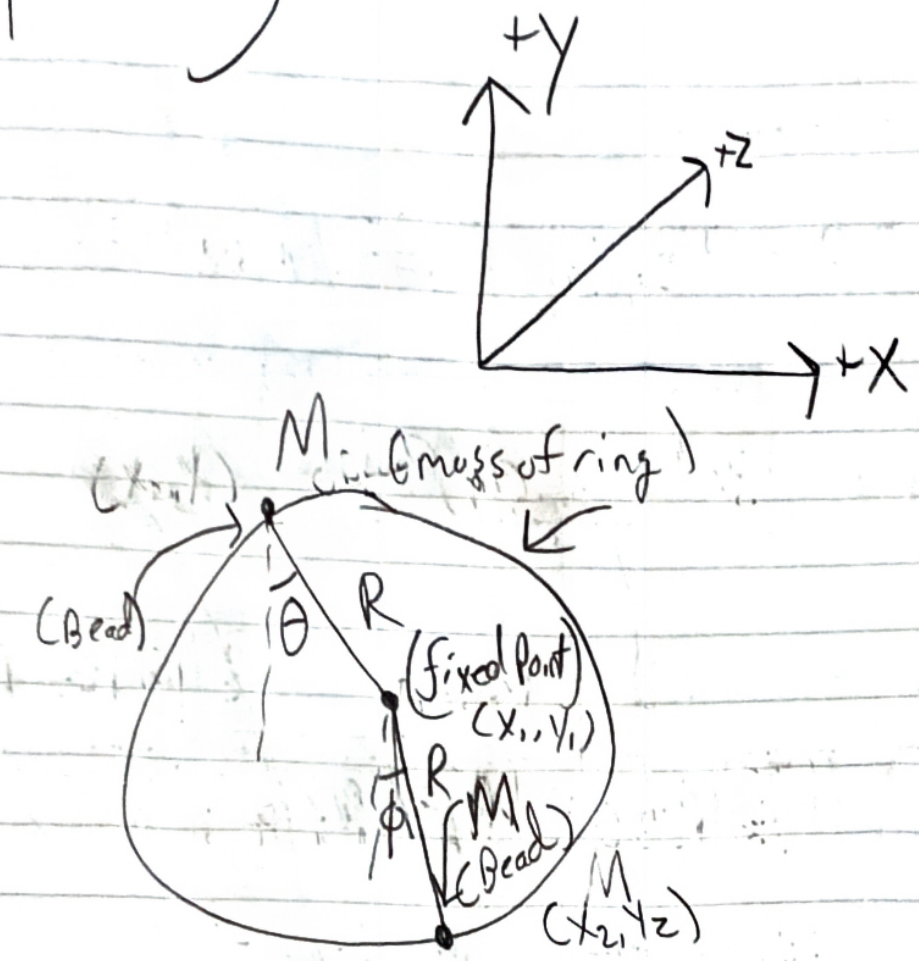
X-equation:

$$X(t) = X_0 \cos(\omega_x t)$$



- X and y frequencies are zero because the motion of these equations represent translational motion but not oscillatory

prob 2.)



- R → radius of hoop
- M → mass of bead
- M → mass of hoop

a.) (X_2, Y_2) → coordinates of bead)

(X_1, Y_1) → coordinate of center of hoop.

∴ N = 2 objects of interest

$Z_1 = 0$, $Z_2 = 0$, $X_1^2 + Y_1^2 = R^2$, $X_2^2 + Y_2^2 = R^2$

$k = 4$

$n = 3(2) - 2 = 2$ degrees of freedom

Center point equations:

- $X_1 = R\sin\theta$
- $Y_1 = R - R\cos\theta = R(1-\cos\theta)$

Bead equations:

- $X_2 = R\sin\theta + R\sin\phi = R(\sin\theta + \sin\phi)$
- $Y_2 = (R - R\cos\theta) + (R - R\cos\phi)$
- $Y_2 = R(1-\cos\theta) + R(1-\cos\phi)$
- $Y_2 = R(2 - \cos\theta - \cos\phi)$

$$\dot{X}_1 = R\cos\theta\dot{\theta}, \quad \dot{Y}_1 = R\sin\theta\dot{\theta}$$

$$\dot{X}_2 = R(\cos\theta\dot{\theta} + \cos\phi\dot{\phi})$$

$$\dot{Y}_2 = R(\sin\theta\dot{\theta} + \sin\phi\dot{\phi})$$

$\theta \rightarrow$ Displacement of hoop

$\phi \rightarrow$ Displacement of bead

- $T = T_{Hoop} + T_{Bead}$
- $T_{Hoop} = T_{trans.} + T_{rotational}$
  - Thin circular loop: $I_{cm} = mr^2$
  - Parallel axis theorm:

$$I_{cm} = MR^2$$

- $T_{Hoop} = \frac{1}{2}MR^2\dot{\theta}^2 + \frac{1}{2}M(\dot{X}_1^2 + \dot{Y}_1^2)$
- $T_{Hoop} = \frac{1}{2}MR^2\dot{\theta}^2 + \frac{1}{2}(R^2\cos^2\theta\dot{\theta}^2 + R\sin^2\theta\dot{\theta}^2)M$
- $T_{Hoop} = \frac{1}{2}MR^2\dot{\theta}^2 + \frac{1}{2}(R^2\dot{\theta}^2)M$
- $T_{Hoop} = MR^2\dot{\theta}^2$
- $T_{Bead} = \frac{1}{2}M(\dot{X}_2^2 + \dot{Y}_2^2) =$

$$\dot{x}_2^2 = R^2\left(\cos^2\theta\dot{\theta}^2 + 2\cos\theta\dot{\theta}\dot{\phi}\cos\phi + \cos^2\phi\dot{\phi}^2\right)$$

$$\dot{y}_2^2 = R^2\left(\sin^2\theta\dot{\theta}^2 + 2\sin\theta\dot{\theta}\dot{\phi}\sin\phi + \sin^2\phi\dot{\phi}^2\right)$$

$$T_{Bead} = \frac{1}{2}MR^2\left(\dot{\theta}^2 + \dot{\phi}^2 + 2\dot{\theta}\dot{\phi}\left(\cos\theta\cos\phi + \sin\theta\sin\phi\right)\right)$$

$$T_{Bead} = \frac{1}{2}MR^2\left(\dot{\theta}^2 + \dot{\phi}^2 + 2\dot{\theta}\dot{\phi}\cos\left(\theta - \phi\right)\right)$$

$$\boxed{T = MR^2\dot{\theta}^2 + \frac{1}{2}MR^2\left(\dot{\theta}^2 + \dot{\phi}^2 + 2\dot{\theta}\dot{\phi}\cos\left(\theta - \phi\right)\right)}$$

$$V = -Mgy_1 - Mgy_2$$

$$\boxed{V = -MgR\left(1 - \cos\theta\right) - MgR\left(2 - \cos\theta - \cos\phi\right)}$$

$$L = MR^2\dot{\theta}^2 + \frac{1}{2}MR^2\left(\dot{\theta}^2 + \dot{\phi}^2 + 2\dot{\theta}\dot{\phi}\cos(\theta-\phi)\right) + MgR(1-\cos\theta) + MgR(2-\cos\theta-\cos\phi)$$

b)

- $\cos\theta \approx 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!}$
- $\cos\phi \approx 1 - \frac{\phi^2}{2!} + \frac{\phi^4}{4!}$
- $\cos(\theta-\phi) \approx 1 - \frac{(\theta-\phi)^2}{2!} + \frac{(\theta-\phi)^4}{4!}$

$$L = MR^2\dot{\theta}^2 + \frac{1}{2}MR^2\left(\dot{\theta}^2 + \dot{\phi}^2 + 2\dot{\theta}\dot{\phi}\right) + MgR\left(1-1+\frac{\theta^2}{2!}\right) + MgR\left(2-1-1+\frac{\phi^2}{2!}\right)$$

$$L = MR^2\dot{\theta}^2 + \frac{1}{2}MR^2\left(\dot{\theta}^2 + \dot{\phi}^2 + 2\dot{\theta}\dot{\phi}\right) + MgR\left(\frac{\theta^2}{2}\right) + MgR\left(\frac{\phi^2}{2}\right)$$

$$\cdot\ T = MR^2\dot{\theta}^2 + \frac{1}{2}MR^2\left(\dot{\theta}^2 + \dot{\phi}^2 + 2\dot{\theta}\dot{\phi}\right)$$

$$\cdot\ 2T = 2M\dot{\theta}^2R^2 + MR^2\left(\dot{\theta}^2 + \dot{\phi}^2 + 2\dot{\theta}\dot{\phi}\right)$$

$$\cdot\ 2T = 2M\dot{\theta}^2R^2 + MR^2\dot{\theta}^2 + MR^2\dot{\phi}^2 + 2MR^2\dot{\theta}\dot{\phi}$$

$$= \begin{pmatrix}\dot{\theta} & \dot{\phi}\end{pmatrix}\begin{pmatrix}3MR^2 & MR^2 \\ MR^2 & MR^2\end{pmatrix}\begin{pmatrix}\dot{\theta} \\ \dot{\phi}\end{pmatrix}$$

$$T_{ij} = \begin{pmatrix}3MR^2 & MR^2 \\ MR^2 & MR^2\end{pmatrix}$$

$$2V = -MgR\left(\theta^2\right) + MgR\left(\phi^2\right)$$

$$2V = -\begin{pmatrix}\theta, \phi\end{pmatrix}\begin{pmatrix}MgR & 0 \\ 0 & MgR\end{pmatrix}\begin{pmatrix}\theta \\ \phi\end{pmatrix}$$

$$V_{ij} = \begin{pmatrix}2MgR & 0 \\ 0 & MgR\end{pmatrix}$$

c.)

$$\left[V - \omega_k^2 T\right] = 0$$

$$V - \omega_k^2 T = \begin{pmatrix} 2MgR - \omega_k^2 3MR^2 & -MR^2\omega_k^2 \\ -MR^2\omega_k^2 & MgR - \omega_k^2 MR^2 \end{pmatrix}$$

$$\det\left(V - \omega_k^2 T\right) = \begin{pmatrix} 2MgR - \omega_k^2 3MR^2 & -MR^2\omega_k^2 \\ -MR^2\omega_k^2 & MgR - MR^2\omega_k^2 \end{pmatrix}$$

$$\left(2MgR - 3MR^2\omega_k^2\right)\left(MgR - MR^2\omega_k^2\right) - \left(MR^2\omega_k^2\right)^2 = 0$$

$$2M^2g^2R^2 - 2M^2R^3g\omega_k^2 - 3M^2R^3g\omega_k^2 + 3M^2R^4\omega_k^4 + M^2R^4\omega_k^4 = 0$$

$$4M^2R^4\omega_k^4 - 5M^2R^3g\omega_k^2 + 2M^2g^2R^2 = 0$$

$$\omega^4 - \frac{5g}{2R}\omega_k^2 + \frac{g^2}{R^2} = 0$$

- $\omega_k^4 - \frac{5g}{2R}\omega_k^2 + \frac{g^2}{R^2} = 0$

- $\omega_k^2 = \omega$

- $\omega^2 - \frac{5g}{2R}\omega + \frac{g^2}{R^2} = 0$

- $b = -\frac{5g}{2R}$, $C = \frac{g^2}{2R^2}$

- $A = 1$

- $\omega = \frac{\frac{5g}{2R} \pm \sqrt{\frac{25g^2}{4R^2} - \frac{4g^2}{R^2}}}{(2)}$

- $\omega = \frac{\frac{5g}{2R} \pm \sqrt{\frac{25}{4}\frac{g^2}{R^2} - \frac{16}{4}\frac{g^2}{R^2}}}{2}$

- $\omega = \frac{\frac{5g}{2R} \pm \sqrt{\frac{9g^2}{4R^2}}}{2} = \frac{5}{2}\frac{g}{R} \pm \frac{3}{2}\frac{g}{R}$

- $\omega = \frac{2}{4}\frac{g}{R}$, $\omega = \frac{1}{2}\frac{g}{R}$

$$w_k^2 = w$$

$$w_1 = \sqrt{\frac{2g}{R}}$$

$$w_2 = \sqrt{\frac{1}{2}\frac{g}{R}}$$

$$[V - w^2 T]\,\vec{a}_k = 0$$

$$w_1^2 = \frac{2g}{R}$$

$$\vec{a}_1 = \begin{pmatrix} a_x \\ a_y \end{pmatrix}$$

$$\left[V - \frac{2g}{R} T\right] \vec{a}_k = 0$$

$$\begin{pmatrix} \overbrace{2MgR - 2g3MR}^{6MgR} & \overbrace{-MR2g}^{-2MgR} \\ -MR^2\frac{2g}{R} & MgR - \frac{2g}{R}MR^2 \end{pmatrix} \begin{pmatrix} a_x \\ a_y \end{pmatrix} = 0$$

$$\begin{pmatrix} -4MgR & -2MgR \\ -2MgR & -MgR \end{pmatrix} \begin{pmatrix} a_x \\ a_y \end{pmatrix} = 0$$

- $-4MgRa_x - 2MgRa_y = 0$
- $-2MgRa_x - MgRa_y = 0$

- $a_x = -\frac{1}{2}a_y$, $a_x = -1/2$, $a_y = 1$

$$\boxed{\vec{q} = \begin{pmatrix} -1/2 \\ 1 \end{pmatrix}}$$

$\omega_2^2 = \frac{g}{2R}$:

$$\vec{a}_2 = \begin{pmatrix} a_x \\ a_y \end{pmatrix}$$

$$\begin{pmatrix} 2MgR - \frac{3MgR}{2} & -\frac{MgR}{2} \\ -\frac{MgR}{2} & MgR - \frac{MgR}{2} \end{pmatrix} \begin{pmatrix} a_x \\ a_y \end{pmatrix} = 0$$

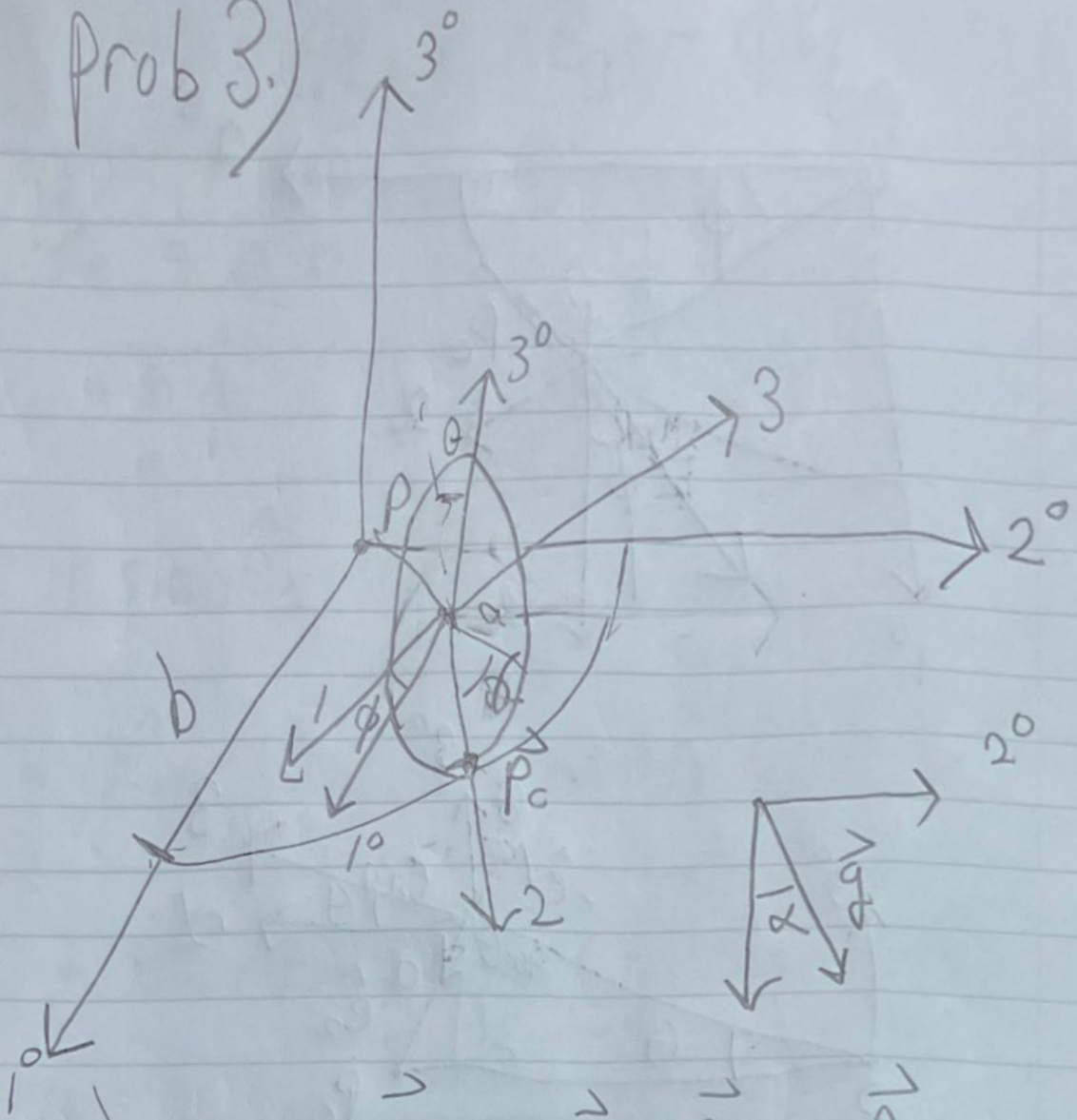
$$\begin{pmatrix} MgR/2 & -MgR/2 \\ -MgR/2 & MgR/2 \end{pmatrix} \begin{pmatrix} a_x \\ a_y \end{pmatrix} = 0$$

$$a_x \frac{MgR}{2} = \frac{MgR}{2} a_y$$

$$a_x = a_y$$

$$\vec{a}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

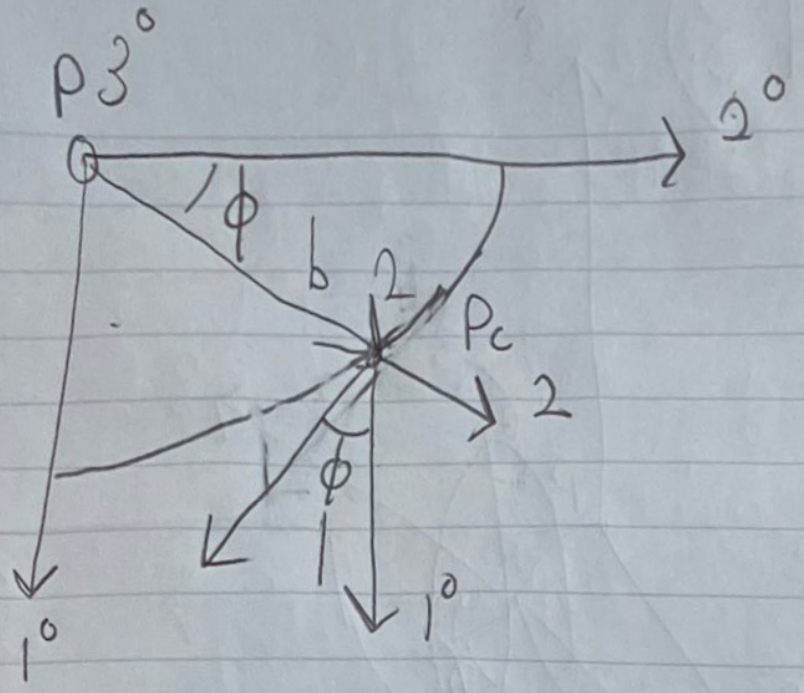
Prob 3.)

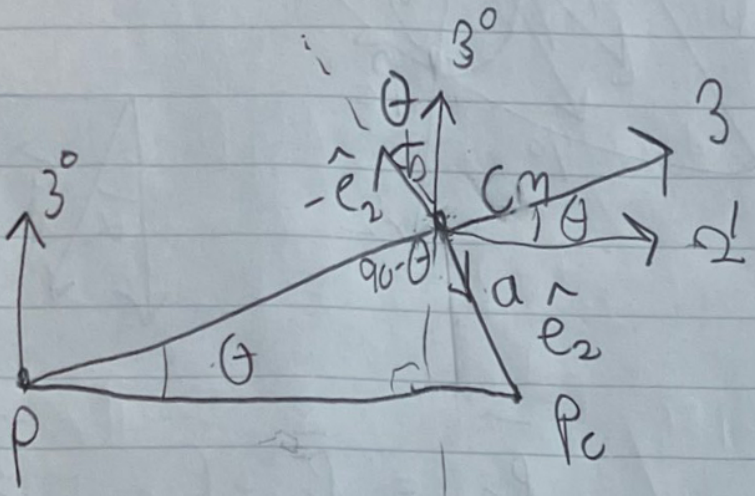


a.)

$$\vec{R}_{cm} = \vec{P}_c + \vec{R}_{cm} - \vec{P}_c$$

- $\hat{e}_3^\circ \leftrightarrow$ perpendicular to inclined plane whose axis $\hat{e}_2^\circ$ points down the plane.





$$\vec{P_c} = b(\cos\phi \hat{e}_2^{\circ} + \sin\phi \hat{e}_1^{\circ})$$

$$R_{cm} = (\cos\phi \hat{e}_2^{\circ} + \sin\phi \hat{e}_1^{\circ})(b - a\sin\theta) + a\cos\theta \hat{e}_3^{\circ}$$

a rad.

- $\vec{\omega} = \dot{\theta}\hat{e}_1 + \dot{\gamma}\hat{e}_3 - \dot{\phi}\hat{e}_3$

- $\dot{\gamma}_a = \dot{\phi}\, b$

- $\dot{\gamma}_a = \frac{b}{a}\dot{\phi}$

- $\hat{e}_3 = \sin\theta\hat{e}_3 - \cos\theta\hat{e}_2$

- $\vec{\omega} = \dot{\theta}\hat{e}_1 + \frac{b}{a}\dot{\phi}\hat{e}_3 - \left(\sin\theta\hat{e}_3 - \cos\theta\hat{e}_2\right)\dot{\phi}$

- $\boxed{\vec{\omega} = \dot{\theta}\hat{e}_1 + \dot{\phi}\cos\theta\hat{e}_2 + \dot{\phi}\left(\frac{b}{a} - \sin\theta\right)\hat{e}_3}$

b) - $T = \frac{1}{2}\left(I_1\omega_1^2 + I_2\omega_2^2 + I_3\omega_3^2\right)$

- $\left(I_1 = I_2\right)$ (because of symmetry 1-2 axis)

$$\boxed{T = \frac{1}{2}I_1\dot{\theta}^2 + \frac{1}{2}I_1\dot{\phi}^2\cos^2\theta + \frac{1}{2}I_3\dot{\phi}^2\left(\frac{b}{a} - \sin\theta\right)^2}$$